Titanic Dataset — Project 2 Documentation

Exploratory Data Analysis (EDA)

DecodeLabs Industrial Training Kit — Data Analytics Project 2

Submitted By

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Project: Exploratory Data Analysis (EDA)

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the cleaned Titanic dataset to identify patterns, trends, distributions, correlations, and important insights.

EDA helps in understanding the dataset and discovering meaningful relationships between variables before applying Machine Learning algorithms.

Dataset Information

Dataset Name

Titanic Dataset

Source

Kaggle Titanic Dataset

Dataset Description

The dataset contains passenger details such as:

- Passenger ID
- Name
- Age
- Gender
- Ticket Fare

- Passenger Class
- Embarkation Port
- Survival Status

The cleaned dataset from Project 1 was used for analysis.

Tools & Technologies Used

Tool	Purpose
Python	Data Analysis
Pandas	Data Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Jupyter Notebook	Project Development

Importing Required Libraries

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns
```

Loading the Cleaned Dataset

```
df = pd.read_csv("cleaned_titanic.csv")
```

Initial Dataset Inspection

Viewing First Five Rows

```
df.head()
```

Checking Dataset Shape

df.shape

Checking Dataset Information

df.info()

Checking Column Names

df.columns

Descriptive Statistics

Descriptive statistics were calculated to understand the dataset.

df.describe()

Mean, Median & Count Analysis

Mean Age

df["Age"].mean()

Median Fare

df["Fare"].median()

Passenger Count

df["PassengerId"].count()

Observation

Most passengers were adults, and ticket fare values varied significantly.

Survival Analysis

Survival Count

df["Survived"].value_counts()

Survival Visualization

sns.countplot(x="Survived", data=df)

plt.title("Survival Count")

plt.show()

Observation

A large number of passengers did not survive.

Gender Analysis

Gender Distribution

```
sns.countplot(x="Sex", data=df)
plt.title("Gender Distribution")
plt.show()
```

Survival Based on Gender

```
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survival Based on Gender")
plt.show()
```

Observation

Female passengers had a higher survival rate compared to male passengers.

Age Distribution Analysis

Histogram of Age

```
plt.hist(df["Age"], bins=20)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```

Observation

Most passengers belonged to the age group of 20–40 years.

Fare Distribution Analysis

Histogram of Fare

```
plt.hist(df["Fare"], bins=30)
```

```
plt.title("Fare Distribution")

plt.xlabel("Fare")

plt.ylabel("Count")

plt.show()
```

Observation

The fare distribution is right-skewed with some extreme values.

Outlier Detection

Boxplot of Fare

```
sns.boxplot(x=df["Fare"])

plt.title("Fare Outliers")

plt.show()
```

Observation

Several outliers were identified in the Fare column.

Passenger Class Analysis

Passenger Class Distribution

```
sns.countplot(x="Pclass", data=df)

plt.title("Passenger Class Distribution")

plt.show()
```

Survival by Passenger Class

```
sns.countplot(x="Pclass", hue="Survived", data=df)

plt.title("Survival by Passenger Class")

plt.show()
```

Observation

Passengers in first class had higher survival rates compared to lower classes.

Correlation Analysis

Correlation Matrix

```
corr = df.corr(numeric_only=True)


corr
```

Correlation Heatmap

```
plt.figure(figsize=(10,6))


sns.heatmap(corr, annot=True)


plt.title("Correlation Heatmap")


plt.show()
```

Observation

Passenger class and fare showed correlation with survival.

Scatter Plot Analysis

Age vs Fare

```
plt.scatter(df["Age"], df["Fare"])


plt.xlabel("Age")
plt.ylabel("Fare")


plt.title("Age vs Fare")


plt.show()
```

Observation

Higher fare passengers were distributed across different age groups.

Embarked Analysis

Embarked Distribution

```
sns.countplot(x="Embarked", data=df)


plt.title("Embarked Distribution")


plt.show()
```

Observation

Most passengers boarded from Southampton.

Missing Value Verification

```
df.isnull().sum()
```

Result

No significant missing values were found in the cleaned dataset.

Key Insights

1. Female passengers had better survival rates.
2. First-class passengers were more likely to survive.
3. Fare distribution contained multiple outliers.
4. Most passengers were young adults.
5. Passenger class strongly influenced survival.
6. Higher ticket fares were associated with higher-class passengers.

Business Understanding

The analysis indicates that socio-economic status played a major role in passenger survival during the Titanic disaster. Passengers with higher ticket classes and higher fares had greater survival chances.

Gender also significantly influenced survival outcomes, as female passengers were prioritized during rescue operations.

Challenges Faced

- Handling skewed fare distribution
- Understanding outlier impact
- Interpreting correlations correctly
- Visualizing multiple variables effectively

Key Learnings

Through this project, the following skills were developed:

- Exploratory Data Analysis (EDA)
- Data Visualization
- Statistical Analysis
- Correlation Analysis
- Outlier Detection
- Data Interpretation
- Business Insight Generation

Conclusion

Exploratory Data Analysis was successfully performed on the Titanic dataset. Important patterns, trends, outliers, and relationships were identified using statistical analysis and visualization techniques.

The project provided valuable insights into passenger survival patterns and improved understanding of analytical workflows used in Data Analytics and Data Science.

Future Scope

The analyzed dataset can now be used for:

- Machine Learning Models

- Survival Prediction
- Classification Algorithms
- Statistical Modeling
- Advanced Data Visualization

Thank You

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